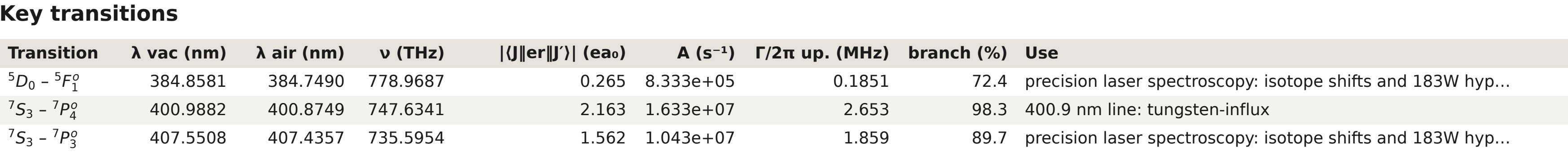
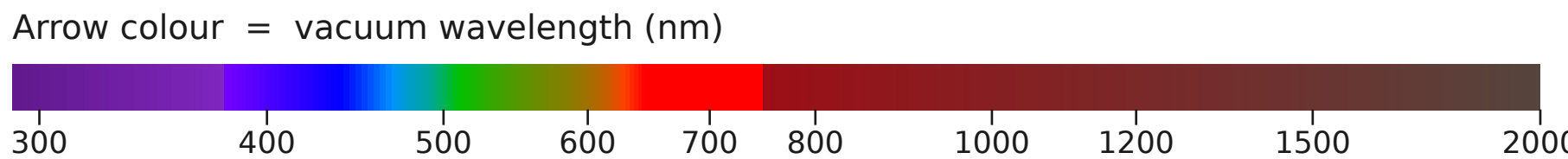


W⁺ ionisation limit 63427.70 cm⁻¹ = 7.86403 eV (157.660 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\{f||e||f'\}$ in $\mathbf{e}_0$
 Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
 Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
 Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



$\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

^{183}W hyperfine structure ($I = 1/2$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
2S_3	505.592(12)	-	F=5/2: -1011.18 F=7/2: +758.39
${}^5F_1^o$	47.1(11)	-	F=1/2: -47.10 F=3/2: +23.55
${}^2P_3^o$	496.2(6)	-	F=5/2: -992.40 F=7/2: +744.30
${}^2P_2^o$	440.5(6)	-	F=7/2: -1101.25 F=9/2: +881.00

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$^7\text{D}_0 - ^7\text{F}_9^o$	384.858	186-184	-12.7(16)
$^7\text{D}_0 - ^7\text{F}_9^o$	384.858	184-182	-10.0(16)
$^7\text{D}_0 - ^7\text{F}_9^o$	384.858	183-182	-4.5(16)
$^7\text{S}_3 - ^7\text{P}_2^o$	400.988	186-184	423.6(21)
$^7\text{S}_3 - ^7\text{P}_2^o$	400.988	184-182	498.4(21)
$^7\text{S}_3 - ^7\text{P}_2^o$	400.988	183-182	263.2(21)
$^7\text{S}_3 - ^7\text{P}_2^o$	400.988	182-180	334.0(21)
$^7\text{S}_3 - ^7\text{P}_2^o$	407.551	186-184	463.6(29)
$^7\text{S}_3 - ^7\text{P}_2^o$	407.551	184-182	541.4(29)
$^7\text{S}_3 - ^7\text{P}_2^o$	407.551	183-182	286.9(29)
$^7\text{S}_3 - ^7\text{P}_2^o$	407.551	182-180	385.6(29)

$$\text{Shift} = \nu(\text{first isotope}) - \nu(\text{second isotope}).$$